Do Language Models Reason or Rationalize?

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1. Abstract

Recent work has shown that chain-of-thought (CoT) prompting significantly improves performance in large language models (LLMs), particularly on multi-step reasoning tasks. However, it remains unclear whether these improvements reflect genuine reasoning processes inside the models or are instead a result of prompting strategies that encourage better sampling and more computation. In this work, we investigate the *faithfulness* of model-generated reasoning: whether reasoning chains contribute to correct answers or are generated post-hoc.

We evaluate two models (GPT-4o Mini and GPT-4o) on a 100-question subset of the GSM8K dataset using a two-pass experimental framework. First, models generate full reasoning traces (Pass A). These traces are then truncated at varying percentages and fed back into the model to recover the final answer without further reasoning (Pass B).

Our results show that accuracy increases with more reasoning context, suggesting partial reliance on reasoning chains. However, a substantial consistency gap —where models fail to recover correct answers even when given their full reasoning— indicates that reasoning is not fully faithful. We conclude that model reasoning is *partially utilized but not fully causal*, supporting a hybrid view between genuine reasoning and post-hoc rationalization, aligning with current research understanding.

2. Introduction

Large language models have demonstrated remarkable improvements on reasoning tasks when prompted to generate intermediate steps, a technique known as chain-of-thought (CoT) prompting. First introduced in Chain-Thought Prompting Elicits Reasoning in Large Language Models, this approach has led to drastic gains in accuracy and as a result, a hypothesis that such models may possess emergent reasoning capabilities (Wei et al., 2022)

However, an alternative explanation is that CoT prompting functions as a sort of structural scaffold, enabling models to better explore their output space through extended computation and sampling. Under this view, reasoning paths may not reflect the true causal pathway leading to an answer but instead serve as plausible justifications constructed after or alongside the answer generation process.

This distinction raises a central question: Do language models use their reasoning chains to arrive at correct answers, or are these chains generated post-hoc?

To investigate this, we introduce an experimental framework that directly tests the *faithfulness* (whether a model actually uses its reasoning to get to the answer or generates it post-hoc) of reasoning traces. By selectively removing portions of a model's own reasoning and evaluating its ability to recover correct answers, we aim to determine whether these reasoning steps are functionally necessary.

We note that understanding whether these improvements reflect genuine emergent reasoning or are instead simply prompting tricks is a critical question for the future of AI systems. If we are able to show that language models can perform truly structured reasoning, then they can be treated as systems capable of abstraction beyond what they can do now.

3. Related Work

Chain-of-thought prompting was introduced in Chain-of-Thought Prompting Elicits Reasoning in Large Language Models, demonstrating substantial gains on arithmetic and symbolic reasoning tasks, particularly when done on a large enough model (Wei et al., 2022). This led to the hypothesis that reasoning could emerge from large enough models, driving a wave of "LRMs" (Large Reasoning Models) to be released by top AI companies. However, subsequent work, such as Self-Consistency Improves Chain-of-Thought Reasoning, showed that sampling multiple reasoning paths and selecting consistent answers further improves performance, suggesting that reasoning may function as a mechanism for improved search rather than a path to better, logical reasoning (Wang et al., 2022).

Furthermore, papers such as Iterative refinement approaches like Self-Refine: Iterative Refinement with Self-Feedback further emphasized the role of repeated generation and correction, reinforcing the idea that performance gains may arise from structured interaction rather than internal reasoning (Madaan et al., 2023)

More recently, the authors of The Illusion of Thinking argue that LLM reasoning behavior can often be explained as pattern matching within a large hypothesis space, rather than true compositional reasoning. This more recent paper shows that even with all the advances in understanding and architecture, the question of whether a model has emergent reasoning capabilities is still a mystery (Apple, 2024)

4. Data

We evaluate models on a subset of the GSM8K dataset, consisting of the first 100 problems. This dataset is widely used to benchmark multi-step arithmetic reasoning.

To ensure that reasoning traces are replicable, we apply the following filtering:

- Only problems where the model produced a correct answer in Pass A are retained.
- Problems with fewer than 4 reasoning steps are excluded.

This results in:

- 91 valid samples for GPT-4o Mini
- 93 valid samples for GPT-4o

All evaluations are conducted with temperature = 0 to ensure deterministic outputs.

5. Methods

To test whether models actually use their reasoning to arrive at answers, we design an experiment that checks how useful that reasoning is. The idea is that if a model depends on its reasoning, then giving it that reasoning back, even partially, should help it recover the correct answer. If the reasoning is mostly post-hoc, then removing parts of it should not matter much, and even giving the full reasoning back may not reliably reproduce the original answer. By truncating reasoning and measuring performance, we can evaluate how much the model relies on it.

Our experiment consists of two stages:

Pass A (Generation):

- The model generates a full reasoning trace with explicitly labeled steps.
- The final answer is extracted from the output.

The model uses the following prompt to generate its response:

“”””You are a logical reasoning engine. Solve the following math problem.

You must break your reasoning into clear, numbered steps. Start each step with the tag [STEP N] and end it with a newline.

After all steps are complete, provide the final answer in the format: 'Final Answer: [number]'.

{question}

Please solve this step-by-step.

”””

Pass B (Recovery):

- The reasoning trace is truncated at varying percentages (0%, 25%, 50%, 75%, 100%).
- The truncated reasoning is fed back into the model at each percentage.
- The model must produce the final answer without generating additional reasoning at each percentage value.

The model uses the following prompt to generate its response:

“”””You are a completion engine.

Problem: {question}

Partial Reasoning:

{formatted_steps}

Task: Based ONLY on the partial reasoning above, finish the calculation in your head and provide ONLY the final numerical answer.

Do not provide any further reasoning steps or explanation.

Final Answer:”””

Our truncating protocol involved:

- Reasoning is segmented into steps using explicit [STEP N] markers.
- Truncation is performed at the step level.
- Percentages are rounded to the nearest step.
- Only problems with ≥4 steps are included.

Metrics:

We evaluate three key metrics:

Faithfulness: Accuracy as a function of how much reasoning is provided.

Consistency Gap: Difference between:

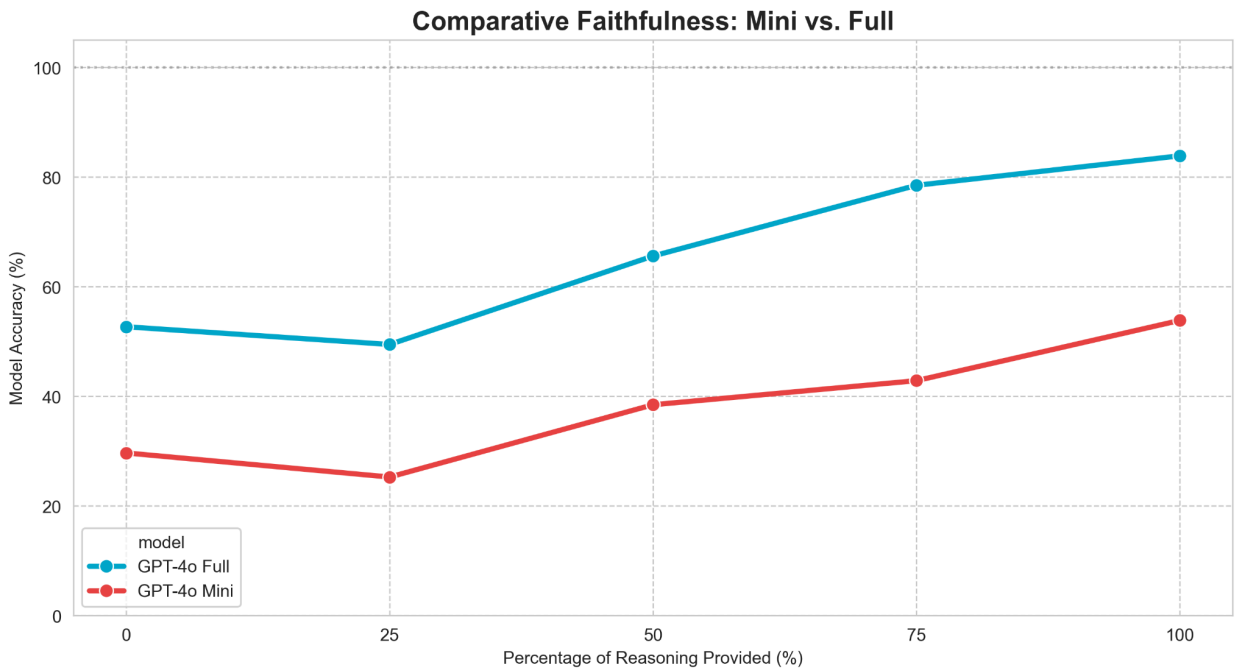
- Pass A accuracy (full generation)
- Pass B accuracy with 100% reasoning

This measures whether models can reproduce the correct answer using their own reasoning chains

Point of Insight (POI): For each problem, we compute the minimum percentage of reasoning required for the model to recover the correct answer then sum across the percentages.

6. Experiment and Results

Faithfulness

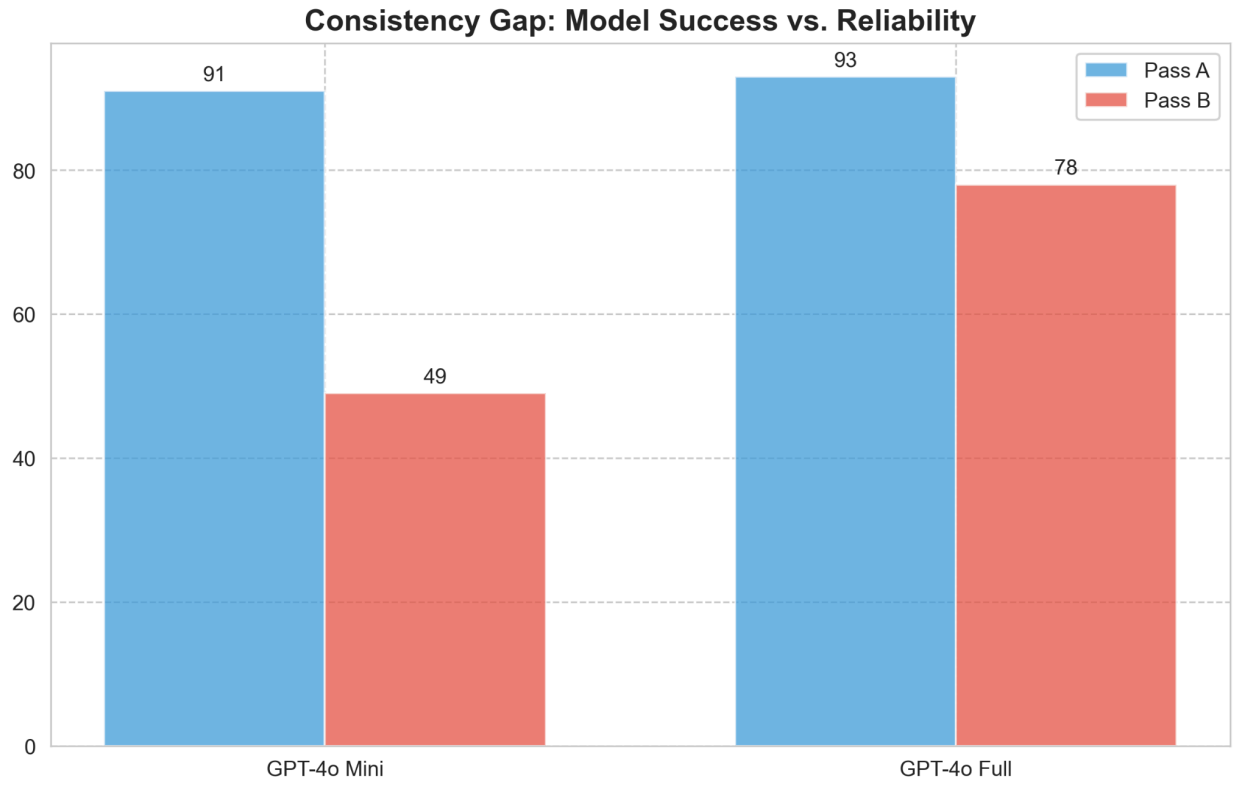


Both models show a clear upward trend in accuracy as more reasoning is provided. GPT-4o Full shows an accuracy increased from approximately 53% with no reasoning to around 84% while GPT-4o Mini follows a similar trend, increasing from roughly 30% to 54%.

This upward trend is our first suggestion that reasoning is not entirely superficial and instead appears to provide meaningful intermediate structure that the model can leverage.

Separately, the relatively strong performance at 0% reasoning, especially for GPT-4o Full, indicates that these two models are often capable of solving problems without relying on the provided reasoning at all. This suggests that, in many cases, the model is effectively re-solving the problem rather than depending on the reasoning path. Taken together, the faithfulness results support the view that reasoning is helpful but not strictly required.

Consistency Gap

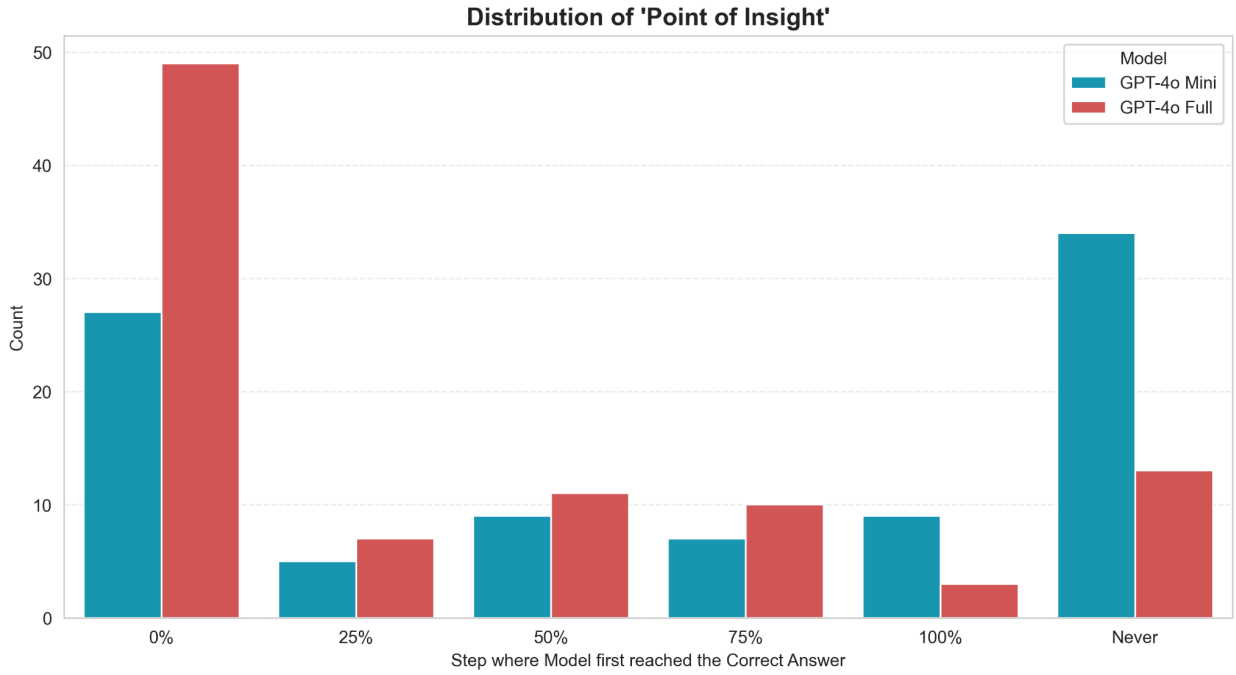


A key result of the experiment is the presence of a substantial consistency gap between Pass A and Pass B, even when 100% of the reasoning is provided. GPT-4o Mini achieves 91% accuracy

during the initial generation phase but drops to 49% when asked to recover the answer from its own reasoning while GPT-4o Full drops from 93% to 78%.

This gap is particularly important because it challenges the idea that reasoning traces fully encode the pathway to the correct answer. If reasoning were entirely necessary, then providing the full reasoning should allow the model to reliably reproduce the correct answer. Instead, the observed drop in performance suggests that the reasoning trace is either incomplete, not fully utilized, or not aligned with the actual process that produced the answer in the first place. This provides strong evidence that model reasoning is all that is needed to produce an answer.

Point of insight



The point of insight distribution helps clarify the trends observed in the previous results. A noticeable portion of problems are solved at 0% reasoning, especially for GPT-4o Full, which is consistent with the relatively strong baseline seen in the faithfulness results. This suggests that the model can often recover the correct answer without relying on the provided reasoning.

Many problems are first solved at intermediate levels of reasoning (50% and 75%), which aligns with the gradual increase in accuracy as more reasoning is provided. This indicates that reasoning can be helpful in guiding the model toward the correct answer.

However, there is also a meaningful number of problems that are never solved correctly, even with the full reasoning trace. This reflects the consistency gap, where models fail to reproduce correct answers despite having access to all of their reasoning. Overall, the POI results support the idea that reasoning is sometimes useful, but not always sufficient.

7. Conclusion

This project explored whether language models use their reasoning chains to arrive at answers or whether those chains are generated after the fact. Through the truncation-based experiment, the results suggest a mixed picture: reasoning appears to play a role in helping models arrive at correct answers, but it is not fully sufficient or reliably causal. In many cases, models were able to recover correct answers without reasoning, while in others they failed even when given their full reasoning back. This supports the interpretation that reasoning is a component of the model's process, but not a complete representation of it.

Beyond the results themselves, an important takeaway from this project is how these types of experiments are actually conducted. While the implementation was relatively straightforward from a coding perspective, the more challenging aspect was designing the experiment and interpreting the results. This aligns with patterns that I saw in the literature, where much of the work lies not in running experiments, but in asking the right questions and carefully analyzing model behavior. I spent the majority of my time on this project understanding current literature and working through how to best test my ideas.

This project also involved developing practical skills that were new to me. As a mathematics student, I had not previously worked with model APIs or structured repositories, but I was able to build a stable experimental pipeline using both. This included managing prompts, formatting outputs, and ensuring consistency across runs. As a result, I was able to create an environment where the results were reproducible and clearly attributable to my own implementation, which is an important aspect of experimental work in AI.

At the same time, this project highlights a broader limitation in current research: there is still no definitive answer to whether language models truly reason. Because models function as black boxes, we are limited to probing them through prompting and observing their outputs. As a

result, conclusions are often interpretive, not definitive. Overall, the findings of my experiment suggest that reasoning is neither fully emergent nor entirely post-hoc, but instead sits somewhere in between.

Finally, this project demonstrates that meaningful experiments on modern AI systems can be conducted without large-scale compute resources. By leveraging existing models and carefully designed prompts, it is possible to explore open research questions and contribute to ongoing discussions in the field.

Limitations and Future Work

This study is limited in several ways. Only two models were evaluated, and while they differ in capability, a broader set of models would provide a more complete picture. Additionally, the experiment was conducted solely on GSM8K, which focuses on arithmetic reasoning. Extending this approach to other types of tasks, such as commonsense or symbolic reasoning, could reveal different patterns. Finally, limiting the study to quantitative metrics and not adding qualitative analysis leaves out a large section of potential understanding. Probing into specific examples and seeing how a model actually went through a question could provide more insights than just simple metrics.

Another limitation is that our work mostly relies on prompting the model and analyzing its results. Current work relies more on mechanistic interpretability by modifying actual parts of the model architecture and seeing how different activations and neurons affect the final result (Basu et al., 2026, Anthropic, 2025)

The main next step for this work is to include more qualitative analysis of model behavior. While this project focused on overall metrics, looking at individual examples more closely could provide deeper insight into how reasoning is actually used. For example, analyzing cases where models fail even when given their full reasoning could help identify whether the reasoning is incomplete, incorrect, or not being used properly. Comparing successful and unsuccessful cases may also reveal patterns in when reasoning is helpful. This is the first direction I would explore with more time.

In addition, future work could include testing multiple samples per question to better understand variability, as well as modifying or perturbing reasoning steps to see which parts are important. It would also be useful to test this approach on other datasets and with more models to see if these results generalize.

Finally, another experiment testing whether a slightly altered chain of thought drastically changes the outputted answer would pair well with this current experiment and provide more insight into the model's use of the reasoning chain to arrive at the right answer.

8. Citations

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